

Yesterday, we learnt a little about the very useful library Pandas widely used for Data Analysis.

✓ It is time to put the many utilities of this library in front of our eyes. We will be performing following manipulations and exploration of our data:

- Indexing and slicing operations
- Manipulating columns
- Exploring unique and missing values

```
# Import pandas package
import pandas as pd
import numpy as np
```

✓ Let us read our data first

```
# loading csv File
file_path = 'imdb_data.csv'
df=pd.read_csv(file_path)
```

```
df.shape
```

```
(3000, 23)
```

```
df.head()
```

	id	belongs_to_collection		budget	genres	homepage	imdb_id	original_language	original_title	overview
0	1	[[{'id': 313576, 'name': 'Hot Tub Time Machine ...		14000000	[[{'id': 35, 'name': 'Comedy'}]]	NaN	tt2637294	en	Hot Tub Time Machine 2	When Lou, who has become the "father of the In...
1	2	[[{'id': 107674, 'name': 'The Princess Diaries ...		40000000	[[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'nam...	NaN	tt0368933	en	The Princess Diaries 2: Royal Engagement	Mia Thermopolis is now a college graduate and ...
2	3		NaN	3300000	[[{'id': 18, 'name': 'Drama'}]]	http://sonyclassics.com/whiplash/	tt2582802	en	Whiplash	Under the direction of a ruthless instructor, ...
3	4		NaN	1200000	[[{'id': 53, 'name': 'Thriller'}, {'id': 18, 'n...	http://kahaanithefilm.com/	tt1821480	hi	Kahaani	Vidya Bagchi (Vidya Balan) arrives in Kolkata ...
4	5		NaN	0	[[{'id': 28, 'name': 'Action'}, {'id': 53, 'nam...	NaN	tt1380152	ko	마린보이	Marine Boy is the story of a former national s...

5 rows × 23 columns

```
df.tail(10)
```

	id	belongs_to_collection	budget	genres	homepage	imdb_id	original_language	original_title	overview
2990	2991	NaN	0	[[{'id': 28, 'name': 'Action'}, {'id': 35, 'name': 'Comedy'}]]	NaN	tt0118663	en	B.A.P.S.	Over the top action comedy
2991	2992	[[{'id': 107469, 'name': 'Save The Last Dance Collection'}]]	13000000	[[{'id': 18, 'name': 'Drama'}, {'id': 10751, 'name': 'Dance'}]]	NaN	tt0206275	en	Save the Last Dance	Midnight girl to the rescue
2992	2993	NaN	1135654	[[{'id': 28, 'name': 'Action'}, {'id': 12, 'name': 'Adventure'}]]	NaN	tt0015400	en	The Thief of Bagdad	Recreation of the classic
2993	2994	NaN	60000000	[[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'name': 'Drama'}]]	http://www.theterminal-themovie.com/	tt0362227	en	The Terminal	Navigation through the airport
2994	2995	NaN	0	[[{'id': 18, 'name': 'Drama'}]]	NaN	tt0105327	en	School Ties	Where the hell is the school
2995	2996	NaN	0	[[{'id': 35, 'name': 'Comedy'}, {'id': 10749, 'name': 'Adventure'}]]	NaN	tt0109403	en	Chasers	Military action and adventure
2996	2997	NaN	0	[[{'id': 18, 'name': 'Drama'}, {'id': 10402, 'name': 'Drama'}]]	NaN	tt2364975	sv	Vi är bäst!	Thriller in Stockholm
2997	2998	NaN	65000000	[[{'id': 80, 'name': 'Crime'}, {'id': 28, 'name': 'Action'}]]	NaN	tt0116908	en	The Long Kiss Goodnight	Sarah's home is the scene
2998	2999	NaN	42000000	[[{'id': 35, 'name': 'Comedy'}, {'id': 10749, 'name': 'Adventure'}]]	http://www.alongcamepolly.com/	tt0343135	en	Along Came Polly	Funny guy's special
2999	3000	NaN	35000000	[[{'id': 53, 'name': 'Thriller'}, {'id': 28, 'name': 'Action'}]]	http://www.abductionthefilm.com/	tt1600195	en	Abduction	A man's unconvincing truth

10 rows × 23 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3000 entries, 0 to 2999
Data columns (total 23 columns):
#   Column              Non-Null Count  Dtype
---  -
0   id                   3000 non-null   int64
1   belongs_to_collection 604 non-null    object
2   budget               3000 non-null   int64
3   genres               2993 non-null   object
```

```

4  homepage          946 non-null  object
5  imdb_id           3000 non-null  object
6  original_language 3000 non-null  object
7  original_title     3000 non-null  object
8  overview           2992 non-null  object
9  popularity         3000 non-null  float64
10 poster_path       2999 non-null  object
11 production_companies 2844 non-null  object
12 production_countries 2945 non-null  object
13 release_date       3000 non-null  object
14 runtime            2998 non-null  float64
15 spoken_languages   2980 non-null  object
16 status             3000 non-null  object
17 tagline            2403 non-null  object
18 title              3000 non-null  object
19 Keywords           2724 non-null  object
20 cast              2987 non-null  object
21 crew              2984 non-null  object
22 revenue            3000 non-null  int64
dtypes: float64(2), int64(3), object(18)
memory usage: 539.2+ KB

```

✓ describe()

This method gives the overall descriptive summary of the dataframe. This works only for numerical columns

```
df.describe()
```



	id	budget	popularity	runtime	revenue
count	3000.000000	3.000000e+03	3000.000000	2998.000000	3.000000e+03
mean	1500.500000	2.253133e+07	8.463274	107.856571	6.672585e+07
std	866.169729	3.702609e+07	12.104000	22.086434	1.375323e+08
min	1.000000	0.000000e+00	0.000001	0.000000	1.000000e+00
25%	750.750000	0.000000e+00	4.018053	94.000000	2.379808e+06
50%	1500.500000	8.000000e+06	7.374861	104.000000	1.680707e+07
75%	2250.250000	2.900000e+07	10.890983	118.000000	6.891920e+07
max	3000.000000	3.800000e+08	294.337037	338.000000	1.519558e+09

✓ Indexing and Slicing

```
df.head()
```

		id	belongs_to_collection	budget	genres	homepage	imdb_id	original_language	original_title	overview
0	1		[[{'id': 313576, 'name': 'Hot Tub Time Machine ...	14000000	[[{'id': 35, 'name': 'Comedy'}]]	NaN	tt2637294	en	Hot Tub Time Machine 2	When Lou, who has become the "father of the In...
1	2		[[{'id': 107674, 'name': 'The Princess Diaries ...	40000000	[[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'nam...	NaN	tt0368933	en	The Princess Diaries 2: Royal Engagement	Mia Thermopolis is now a college graduate and ...
2	3		NaN	3300000	[[{'id': 18, 'name': 'Drama'}]]	http://sonyclassics.com/whiplash/	tt2582802	en	Whiplash	Under the direction of a ruthless instructor, ...
3	4		NaN	1200000	[[{'id': 53, 'name': 'Thriller'}, {'id': 18, 'n...	http://kahaanithefilm.com/	tt1821480	hi	Kahaani	Vidya Bagchi (Vidya Balan) arrives in Kolkata ...
4	5		NaN	0	[[{'id': 28, 'name': 'Action'}, {'id': 53, 'nam...	NaN	tt1380152	ko	마린보이	Marine Boy is the story of a former national s...

5 rows × 23 columns

✓ **iloc** gets rows (or columns) at particular positions in the index (so it only takes integers).

```
# Let us get a subset which consists of first 8 rows and first 4 columns
df.iloc[:5,:4]
```

		id	belongs_to_collection	budget	genres
0	1	[[{'id': 313576, 'name': 'Hot Tub Time Machine ...		14000000	[[{'id': 35, 'name': 'Comedy'}]]
1	2	[[{'id': 107674, 'name': 'The Princess Diaries ...		40000000	[[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'nam...
2	3		NaN	3300000	[[{'id': 18, 'name': 'Drama'}]]
3	4		NaN	1200000	[[{'id': 53, 'name': 'Thriller'}, {'id': 18, 'n...
4	5		NaN	0	[[{'id': 28, 'name': 'Action'}, {'id': 53, 'nam...

✓ **loc** gets rows (or columns) with particular labels from the index.

```
# Now let us subset our dataframe in which we want to have first 8 rows identified with their rows labels and some named columns
df.loc[50:55,['budget','genres','runtime']]
```

		budget	genres	runtime
50		560000	[[{'id': 10749, 'name': 'Romance'}, {'id': 18, ...	102.0
51		12000000	[[{'id': 12, 'name': 'Adventure'}, {'id': 35, '...	160.0
52		0	[[{'id': 12, 'name': 'Adventure'}, {'id': 10751...	97.0
53		0	[[{'id': 18, 'name': 'Drama'}, {'id': 10751, 'n...	98.0
54		20000000	[[{'id': 28, 'name': 'Action'}]]	107.0
55		0	[[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'nam...	95.0

```
df.loc[:,['budget','genres','runtime']]
```



	budget	genres	runtime
0	14000000	[{'id': 35, 'name': 'Comedy'}]	93.0
1	40000000	[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'name': 'Drama'}]	113.0
2	33000000	[{'id': 18, 'name': 'Drama'}]	105.0
3	12000000	[{'id': 53, 'name': 'Thriller'}, {'id': 18, 'name': 'Drama'}]	122.0
4	0	[{'id': 28, 'name': 'Action'}, {'id': 53, 'name': 'Thriller'}]	118.0
...	...	...	...
2995	0	[{'id': 35, 'name': 'Comedy'}, {'id': 10749, 'name': 'Comedy'}]	102.0
2996	0	[{'id': 18, 'name': 'Drama'}, {'id': 10402, 'name': 'Drama'}]	102.0
2997	65000000	[{'id': 80, 'name': 'Crime'}, {'id': 28, 'name': 'Action'}]	120.0
2998	42000000	[{'id': 35, 'name': 'Comedy'}, {'id': 10749, 'name': 'Comedy'}]	90.0
2999	35000000	[{'id': 53, 'name': 'Thriller'}, {'id': 28, 'name': 'Action'}]	106.0

3000 rows x 3 columns

df[0:10]

	id	belongs_to_collection	budget	genres	homepage	imdb_id	original_language	original_title	
0	1	[[{'id': 313576, 'name': 'Hot Tub Time Machine ...	14000000	[[{'id': 35, 'name': 'Comedy'}]]	NaN	tt2637294	en	Hot Tub Time Machine 2	b
1	2	[[{'id': 107674, 'name': 'The Princess Diaries ...	40000000	[[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'nam...	NaN	tt0368933	en	The Princess Diaries 2: Royal Engagement	Tt
2	3	NaN	3300000	[[{'id': 18, 'name': 'Drama'}]]	http://sonyclassics.com/whiplash/	tt2582802	en	Whiplash	c in
3	4	NaN	1200000	[[{'id': 53, 'name': 'Thriller'}, {'id': 18, 'n...	http://kahaanithefilm.com/	tt1821480	hi	Kahaani	
4	5	NaN	0	[[{'id': 28, 'name': 'Action'}, {'id': 53, 'nam...	NaN	tt1380152	ko	마린보이	M i (n
5	6	NaN	8000000	[[{'id': 16, 'name': 'Animation'}, {'id': 12, '...	NaN	tt0093743	en	Pinocchio and the Emperor of the Night	t
6	7	NaN	14000000	[[{'id': 27, 'name': 'Horror'}, {'id': 53, 'nam...	http://www.thepossessionmovie.com/	tt0431021	en	The Possession	A a
7	8	NaN	0	[[{'id': 99, 'name': 'Documentary'}]]	NaN	tt0391024	en	Control Room	/ ra
8	9	[[{'id': 256377, 'name': 'The Muppet Collection...	0	[[{'id': 28, 'name': 'Action'}, {'id': 35, 'nam...	NaN	tt0117110	en	Muppet Treasure Island	/ t
9	10	NaN	6000000	[[{'id': 35, 'name': 'Comedy'}, {'id': 10402, '...	NaN	tt0310281	en	A Mighty Wind	Ir C

10 rows × 23 columns

df[['original_language', 'release_date', 'cast']]

	original_language	release_date	cast
0	en	2/20/15	[{'cast_id': 4, 'character': 'Lou', 'credit_id': ...
1	en	8/6/04	[{'cast_id': 1, 'character': 'Mia Thermopolis'...
2	en	10/10/14	[{'cast_id': 5, 'character': 'Andrew Neimann', ...
3	hi	3/9/12	[{'cast_id': 1, 'character': 'Vidya Bagchi', '...
4	ko	2/5/09	[{'cast_id': 3, 'character': 'Chun-soo', 'cred...
...	...	...	...
2995	en	4/22/94	[{'cast_id': 2, 'character': 'Rock Reilly', 'c...
2996	sv	3/28/13	[{'cast_id': 5, 'character': 'Bobo', 'credit_i...
2997	en	10/11/96	[{'cast_id': 10, 'character': 'Samantha Caine ...
2998	en	1/16/04	[{'cast_id': 8, 'character': 'Reuben Feffer', ...
2999	en	9/22/11	[{'cast_id': 2, 'character': 'Nathan Harper', ...

3000 rows x 3 columns

df.columns

```
Index(['id', 'belongs_to_collection', 'budget', 'genres', 'homepage',
      'imdb_id', 'original_language', 'original_title', 'overview',
      'popularity', 'poster_path', 'production_companies',
      'production_countries', 'release_date', 'runtime', 'spoken_languages',
      'status', 'tagline', 'title', 'Keywords', 'cast', 'crew', 'revenue'],
      dtype='object')
```

✓ In fact we can also perform boolean indexing on our dataframe

```
df['original_language'] == 'en'
```

	original_language
0	True
1	True
2	True
3	False
4	False
...	...
2995	True
2996	False
2997	True
2998	True
2999	True

3000 rows x 1 columns

```
# Boolean Indexing
english_movies = df[df['original_language'] == 'en']
english_movies.head()
```



	id	belongs_to_collection		budget	genres	homepage	imdb_id	original_language	original_title	overview
0	1	[[{'id': 313576, 'name': 'Hot Tub Time Machine ...		14000000	[{'id': 35, 'name': 'Comedy'}]	NaN	tt2637294	en	Hot Tub Time Machine 2	When L who l become "fathe the l
1	2	[[{'id': 107674, 'name': 'The Princess Diaries ...		40000000	[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'nam...	NaN	tt0368933	en	The Princess Diaries 2: Royal Engagement	Thermop is no' colle gradu an
2	3		NaN	3300000	[{'id': 18, 'name': 'Drama'}]	http://sonyclassics.com/whiplash/	tt2582802	en	Whiplash	Under directio a ruthl instructor
5	6		NaN	8000000	[{'id': 16, 'name': 'Animation'}, {'id': 12, '...	NaN	tt0093743	en	Pinocchio and the Emperor of the Night	Pinocc and friend: glow wc and a r
6	7		NaN	14000000	[{'id': 27, 'name': 'Horror'}, {'id': 53, 'nam...	http://www.thepossessionmovie.com/	tt0431021	en	The Possession	A young buys antique l at a y si

5 rows × 23 columns



english_movies.shape

 (2575, 23)

```
non_english_movies = df[df['original_language'] != 'en']
non_english_movies.head()
```


	id	belongs_to_collection	budget	genres	homepage	imdb_id	original_language	original_title
2969	2970	NaN	19800000	[{'id': 18, 'name': 'Drama'}, {'id': 9648, 'name': 'na...'}]	NaN	tt0280707	en	Gosford Park
2970	2971	NaN	25000000	[{'id': 53, 'name': 'Thriller'}, {'id': 28, 'name': 'n...'}]	NaN	tt0103855	en	The Bodyguard
2980	2981	NaN	300000	[{'id': 28, 'name': 'Action'}, {'id': 35, 'name': 'nam...'}]	https://www.facebook.com/Actionherobiju/	tt5320514	ml	ആക്ഷൻ ഹീറോ ബിജു
2992	2993	NaN	1135654	[{'id': 28, 'name': 'Action'}, {'id': 12, 'name': 'nam...'}]	NaN	tt0015400	en	The Thief of Bagdad
2993	2994	NaN	60000000	[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'name': 'nam...'}]	http://www.theterminal-themovie.com/	tt0362227	en	The Terminal

5 rows × 23 columns

(df['original_language']=='en') & (df['runtime'] > 120)

	0
0	False
1	False
2	False
3	False
4	False
...	...
2995	False
2996	False
2997	False
2998	False
2999	False

3000 rows × 1 columns

```
long_english_movies = df[(df['original_language']=='en') & (df['runtime'] > 120)]
long_english_movies.head()
```

	id	belongs_to_collection	budget	genres	homepage	imdb_id	original_language	original_title	overview
12	13	NaN	15000000	[{'id': 18, 'name': 'Drama'}]	http://www.dreamworks.com/ab/	tt0169547	en	American Beauty	Lester Burnham, a depressed suburban father in...
14	15	NaN	102000000	[{'id': 28, 'name': 'Action'}, {'id': 53, 'name': 'Sci-Fi'}]	NaN	tt0181689	en	Minority Report	John Anderton is a top 'Precrime' cop in the I...
20	21	NaN	80000000	[{'id': 12, 'name': 'Adventure'}, {'id': 10751, 'name': 'Mystery'}]	NaN	tt0120738	en	Lost in Space	The prospects for continuing life on Earth in ...
33	34	[{'id': 207632, 'name': 'The ABCs of Death Collection'}]	0	[{'id': 27, 'name': 'Horror'}]	NaN	tt2926810	en	ABCs of Death 2	Taking all that was great from the first insta...
37	38	NaN	50000000	[{'id': 28, 'name': 'Action'}, {'id': 18, 'name': 'Drama'}]	NaN	tt0114069	en	Outbreak	A deadly airborne virus finds its way into the...

5 rows × 23 columns



long_english_movies.shape

(506, 23)

Next let us see how we can manipulate columns

df.shape

(3000, 23)

Let's say if you wanted a new column whose values were half the movie runtime

df['runtime']



	runtime
0	93.0
1	113.0
2	105.0
3	122.0
4	118.0
...	...
2995	102.0
2996	102.0
2997	120.0
2998	90.0
2999	106.0

3000 rows × 1 columns



```
# Add new column which is just half the runtime
df['half_runtime'] = df['runtime'] * 0.5
```

```
df.head()
```



	id	belongs_to_collection	budget	genres	homepage	imdb_id	original_language	original_title	overview
0	1	[[{"id": 313576, "name": "Hot Tub Time Machine ..."}]]	14000000	[{"id": 35, "name": "Comedy"}]	NaN	tt2637294	en	Hot Tub Time Machine 2	When Lou, who has become the "father of the In...
1	2	[[{"id": 107674, "name": "The Princess Diaries ..."}]]	40000000	[{"id": 35, "name": "Comedy"}, {"id": 18, "name": "Drama"}]	NaN	tt0368933	en	The Princess Diaries 2: Royal Engagement	Mia Thermopolis is now a college graduate and ...
2	3	NaN	3300000	[{"id": 18, "name": "Drama"}]	http://sonyclassics.com/whiplash/	tt2582802	en	Whiplash	Under the direction of a ruthless instructor, ...
3	4	NaN	1200000	[{"id": 53, "name": "Thriller"}, {"id": 18, "name": "Drama"}]	http://kahaanithefilm.com/	tt1821480	hi	Kahaani	Vidya Bagchi (Vidya Balan) arrives in Kolkata ...
4	5	NaN	0	[{"id": 28, "name": "Action"}, {"id": 53, "name": "Thriller"}]	NaN	tt1380152	ko	마린보이	Marine Boy is the story of a former national s...

5 rows × 24 columns



✓ We can also create a new column by adding two existing columns

```
# Creating another new column by subtracting/adding two existing columns
df['movie_profit'] = df['revenue'] - df['budget']
```

```
df.head()
```



	id	belongs_to_collection	budget	genres	homepage	imdb_id	original_language	original_title	overview
0	1	[[{'id': 313576, 'name': 'Hot Tub Time Machine ...	14000000	[{'id': 35, 'name': 'Comedy'}]	NaN	tt2637294	en	Hot Tub Time Machine 2	When Lou, who has become the "father of the In...
1	2	[[{'id': 107674, 'name': 'The Princess Diaries ...	40000000	[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'nam...	NaN	tt0368933	en	The Princess Diaries 2: Royal Engagement	Mia Thermopolis is now a college graduate and ...
2	3	NaN	3300000	[{'id': 18, 'name': 'Drama'}]	http://sonyclassics.com/whiplash/	tt2582802	en	Whiplash	Under the direction of a ruthless instructor, ...
3	4	NaN	1200000	[{'id': 53, 'name': 'Thriller'}, {'id': 18, 'n...	http://kahaanithefilm.com/	tt1821480	hi	Kahaani	Vidya Bagchi (Vidya Balan) arrives in Kolkata ...
4	5	NaN	0	[{'id': 28, 'name': 'Action'}, {'id': 53, 'nam...	NaN	tt1380152	ko	마린보이	Marine Boy is the story of a former national s...

5 rows × 25 columns



✓ **Next, we can also drop some columns which we want to drop**

```
# Remove column
df.drop(['id'], axis = 1, inplace=True)

df.head()
```

	belongs_to_collection	budget	genres	homepage	imdb_id	original_language	original_title	overview	popu
0	[[{'id': 313576, 'name': 'Hot Tub Time Machine ...	14000000	[[{'id': 35, 'name': 'Comedy'}]]	NaN	tt2637294	en	Hot Tub Time Machine 2	When Lou, who has become the "father of the In...	6.
1	[[{'id': 107674, 'name': 'The Princess Diaries ...	40000000	[[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'nam...	NaN	tt0368933	en	The Princess Diaries 2: Royal Engagement	Mia Thermopolis is now a college graduate and ...	8.
2	NaN	3300000	[[{'id': 18, 'name': 'Drama'}]]	http://sonyclassics.com/whiplash/	tt2582802	en	Whiplash	Under the direction of a ruthless instructor, ...	64.
3	NaN	1200000	[[{'id': 53, 'name': 'Thriller'}, {'id': 18, 'n...	http://kahaanithefilm.com/	tt1821480	hi	Kahaani	Vidya Bagchi (Vidya Balan) arrives in Kolkata ...	3.
4	NaN	0	[[{'id': 28, 'name': 'Action'}, {'id': 53, 'nam...	NaN	tt1380152	ko	마린보이	Marine Boy is the story of a former national s...	1.

5 rows × 24 columns

df.describe()

	budget	popularity	runtime	revenue	half_runtime	movie_profit
count	3.000000e+03	3000.000000	2998.000000	3.000000e+03	2998.000000	3.000000e+03
mean	2.253133e+07	8.463274	107.856571	6.672585e+07	53.928286	4.419452e+07
std	3.702609e+07	12.104000	22.086434	1.375323e+08	11.043217	1.123274e+08
min	0.000000e+00	0.000001	0.000000	1.000000e+00	0.000000	-1.074474e+08
25%	0.000000e+00	4.018053	94.000000	2.379808e+06	47.000000	1.795200e+04
50%	8.000000e+06	7.374861	104.000000	1.680707e+07	52.000000	6.121122e+06
75%	2.900000e+07	10.890983	118.000000	6.891920e+07	59.000000	4.143293e+07
max	3.800000e+08	294.337037	338.000000	1.519558e+09	169.000000	1.316249e+09

✓ You can drop multiple columns at once too.

```
df.drop(['half_runtime', 'poster_path'],axis=1,inplace=True)
```

```
df.drop(['crew'], axis = 1, inplace=True)
```

```
df.head()
```

	belongs_to_collection	budget	genres	homepage	imdb_id	original_language	original_title	overview	popu
0	[[{'id': 313576, 'name': 'Hot Tub Time Machine ...	14000000	[[{'id': 35, 'name': 'Comedy'}]]	NaN	tt2637294	en	Hot Tub Time Machine 2	When Lou, who has become the "father of the In...	6.
1	[[{'id': 107674, 'name': 'The Princess Diaries ...	40000000	[[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'nam...	NaN	tt0368933	en	The Princess Diaries 2: Royal Engagement	Mia Thermopolis is now a college graduate and ...	8.
2	NaN	3300000	[[{'id': 18, 'name': 'Drama'}]]	http://sonyclassics.com/whiplash/	tt2582802	en	Whiplash	Under the direction of a ruthless instructor, ...	64.
3	NaN	1200000	[[{'id': 53, 'name': 'Thriller'}, {'id': 18, 'n...	http://kahaanithefilm.com/	tt1821480	hi	Kahaani	Vidya Bagchi (Vidya Balan) arrives in Kolkata ...	3.
4	NaN	0	[[{'id': 28, 'name': 'Action'}, {'id': 53, 'nam...	NaN	tt1380152	ko	마린보이	Marine Boy is the story of a former national s...	1.

5 rows × 21 columns

✓ We can do a lot of basic operations as well

```
# Find the sum of values in a column
df['budget'].sum()
```

```
np.int64(67594002330)
```

```
# What if there is a null value
df.loc[:, 'revenue'] = np.nan
```

```
df.iloc[:5]
```



homepage	imdb_id	original_language	original_title	overview	popu.
NaN	tt2637294	en	Hot Tub Time Machine 2	When Lou, who has become the "father of the In...	6.5
NaN	tt0368933	en	The Princess Diaries 2: Royal Engagement	Mia Thermopolis is now a college graduate and ...	8.1
'whiplash/	tt2582802	en	Whiplash	Under the direction of a ruthless instructor, ...	64.1
'film.com/	tt1821480	hi	Kahaani	Vidya Bagchi (Vidya Balan) arrives in Kolkata ...	3.1
NaN	tt1380152	ko	마린보이	Marine Boy is the story of a former national s...	1.1



```
df.set_index('imdb_id', inplace=True)
```

```
df.head()
```



imdb_id	belongs_to_collection	budget	genres	homepage	original_language	original_title	overview	popular
tt2637294	[{'id': 313576, 'name': 'Hot Tub Time Machine ...	14000000	[{'id': 35, 'name': 'Comedy'}]	NaN	en	Hot Tub Time Machine 2	When Lou, who has become the "father of the In...	6.57
tt0368933	[{'id': 107674, 'name': 'The Princess Diaries ...	40000000	[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'nam...	NaN	en	The Princess Diaries 2: Royal Engagement	Mia Thermopolis is now a college graduate and ...	8.24
tt2582802	NaN	3300000	[{'id': 18, 'name': 'Drama'}]	http://sonyclassics.com/whiplash/	en	Whiplash	Under the direction of a ruthless instructor, ...	64.29
tt1821480	NaN	1200000	[{'id': 53, 'name': 'Thriller'}, {'id': 18, 'n...	http://kahaanithefilm.com/	hi	Kahaani	Vidya Bagchi (Vidya Balan) arrives in Kolkata ...	3.17
tt1380152	NaN	0	[{'id': 28, 'name': 'Action'}, {'id': 53, 'nam...	NaN	ko	마린보이	Marine Boy is the story of a former national s...	1.14




```
df.iloc[:3]
```



imdb_id	belongs_to_collection	budget	genres	homepage	original_language	original_title	overview	popularity
tt2637294	[[{'id': 313576, 'name': 'Hot Tub Time Machine ...	14000000	[[{'id': 35, 'name': 'Comedy'}]]	NaN	en	Hot Tub Time Machine 2	When Lou, who has become the "father of the In...	6.57
tt0368933	[[{'id': 107674, 'name': 'The Princess Diaries ...	40000000	[[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'nam...	NaN	en	The Princess Diaries 2: Royal Engagement	Mia Thermopolis is now a college graduate and ...	8.24
tt2582802	NaN	3300000	[[{'id': 18, 'name': 'Drama'}]]	http://sonyclassics.com/whiplash/	en	Whiplash	Under the direction of a ruthless instructor, ...	64.29

```
df.loc[['tt2637294', 'tt0368933']]
```



imdb_id	belongs_to_collection	budget	genres	homepage	original_language	original_title	overview	popularity	production_co
tt2637294	[[{'id': 313576, 'name': 'Hot Tub Time Machine ...	14000000	[[{'id': 35, 'name': 'Comedy'}]]	NaN	en	Hot Tub Time Machine 2	When Lou, who has become the "father of the In...	6.575393	[[{'name': 'Pa Pictures', 'id': 4
tt0368933	[[{'id': 107674, 'name': 'The Princess Diaries ...	40000000	[[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'nam...	NaN	en	The Princess Diaries 2: Royal Engagement	Mia Thermopolis is now a college graduate and ...	8.248895	[[{'name': 'Wal Pictures

```
df.head()
```

	belongs_to_collection		budget	genres	homepage	original_language	original_title	overview	popular
imdb_id									
tt2637294	[[{'id': 313576, 'name': 'Hot Tub Time Machine ...		14000000	[[{'id': 35, 'name': 'Comedy'}]]	NaN	en	Hot Tub Time Machine 2	When Lou, who has become the "father of the In...	6.57
tt0368933	[[{'id': 107674, 'name': 'The Princess Diaries ...		40000000	[[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'name': 'Romance'}]]	NaN	en	The Princess Diaries 2: Royal Engagement	Mia Thermopolis is now a college graduate and ...	8.24
tt2582802		NaN	3300000	[[{'id': 18, 'name': 'Drama'}]]	http://sonyclassics.com/whiplash/	en	Whiplash	Under the direction of a ruthless instructor, ...	64.29
tt1821480		NaN	1200000	[[{'id': 53, 'name': 'Thriller'}, {'id': 18, 'name': 'Drama'}]]	http://kahaanithefilm.com/	hi	Kahaani	Vidya Bagchi (Vidya Balan) arrives in Kolkata ...	3.17
tt1380152		NaN	0	[[{'id': 28, 'name': 'Action'}, {'id': 53, 'name': 'Thriller'}]]	NaN	ko	마린보이	Marine Boy is the story of a former national s...	1.14

```
df['runtime'].sum()
```

```
np.float64(323354.0)
```

```
print(f" The shortest movie length is {df['runtime'].min()}")
print(f" The largest movie length is {df['runtime'].max()}")
```

```
The shortest movie length is 0.0
The largest movie length is 338.0
```

```
df[df['runtime']== 0.0]
```

	belongs_to_collection	budget	genres	homepage	original_language	original_title	overview	popularity
imdb_id								
tt2550838	NaN	6843500	[{'id': 35, 'name': 'Comedy'}]	NaN	it	Il peggior Natale della mia vita	NaN	3.80007
tt0768690	NaN	0	[{'id': 18, 'name': 'Drama'}, {'id': 35, 'name': 'Comedy'}]	NaN	ru	А поутру они проснулись	NaN	0.40236
tt2604346	NaN	0	[{'id': 35, 'name': 'Comedy'}]	NaN	es	¿Quién mató a Bambi?	It tells the story of two young friends who mu...	1.92682
tt2076251	NaN	11000000	[{'id': 35, 'name': 'Comedy'}]	NaN	it	La peggior settimana della mia vita	NaN	5.01056
tt0072785	NaN	0	[{'id': 37, 'name': 'Western'}]	NaN	it	Cipolla Colt	Onion Jack (Franco Nero) has bought a piece of...	1.62344
tt3805180	NaN	750000	[{'id': 80, 'name': 'Crime'}, {'id': 35, 'name': 'Comedy'}]	NaN	ru	Все и сразу	NaN	0.20158
tt1133617	NaN	0	[{'id': 80, 'name': 'Crime'}, {'id': 18, 'name': 'Drama'}]	http://www.vt4.be/vermist	en	Vermist	Five members of the Missing Persons Task Force...	0.22923
tt0477337	NaN	5000000	[{'id': 28, 'name': 'Action'}, {'id': 10749, 'name': 'Mystery'}]	NaN	ru	Mechenosets	NaN	0.41479
tt1620549	[{'id': 106790, 'name': 'Hooked Collection', 'poster_path': 'https://image.tmdb.org/t/p/w1280/106790.jpg'}]	3500000	[{'id': 28, 'name': 'Action'}, {'id': 878, 'name': 'Adventure'}]	NaN	ru	На Игре 2. Новyy Уровень	The continuation of the adventures of the hero...	0.88424
tt3956312	NaN	0	[{'id': 10749, 'name': 'Romance'}]	NaN	en	同桌的妳	In the summer of 1993, timid and shy Zhou Xiao...	0.50400
tt0278675	NaN	0	[{'id': 14, 'name': 'Fantasy'}, {'id': 27, 'name': 'Mystery'}]	NaN	fr	Revelation	Since A.D. 50, a mysterious sacred artifact kn...	0.62509
tt2456720	[{'id': 188197, 'name': 'Cetto La Qualunque - ...'}]	5579750	[{'id': 35, 'name': 'Comedy'}]	NaN	it	Tutto tutto niente niente	NaN	2.20890

✓ **apply()** method applies a function to every row in a pandas dataframe

```
# Suppose we want to convert the column network to uppercase
```

```
def to_uppercase(column):
    return column.upper()
```

```
df['Capitalized_title'] = df['title'].apply(to_uppercase)
```

```
df.head()
```



imdb_id	belongs_to_collection	budget	genres	homepage	original_language	original_title	overview	popularity
tt2637294	[[{'id': 313576, 'name': 'Hot Tub Time Machine ...	14000000	[[{'id': 35, 'name': 'Comedy'}]]	NaN	en	Hot Tub Time Machine 2	When Lou, who has become the "father of the In...	6.57
tt0368933	[[{'id': 107674, 'name': 'The Princess Diaries ...	40000000	[[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'name': 'Romance'}]]	NaN	en	The Princess Diaries 2: Royal Engagement	Mia Thermopolis is now a college graduate and ...	8.24
tt2582802	NaN	3300000	[[{'id': 18, 'name': 'Drama'}]]	http://sonyclassics.com/whiplash/	en	Whiplash	Under the direction of a ruthless instructor, ...	64.29
tt1821480	NaN	1200000	[[{'id': 53, 'name': 'Thriller'}, {'id': 18, 'name': 'Drama'}]]	http://kahaanithemovie.com/	hi	Kahaani	Vidya Bagchi (Vidya Balan) arrives in Kolkata ...	3.17
tt1380152	NaN	0	[[{'id': 28, 'name': 'Action'}, {'id': 53, 'name': 'Thriller'}]]	NaN	ko	마린보이	Marine Boy is the story of a former national s...	1.14

5 rows × 21 columns

✓ Another way to use apply() involves the use of lambda() function.

- A lambda() function is a small anonymous function.
- A lambda() function can take any number of arguments, but can only have one expression.

```
# lambda function - Row wise operation
def revised_profit(revenue, budget):
    if revenue > 0:
        new_profit = revenue - budget
    else:
        new_profit = np.nan
    return new_profit
```

```
df['new_profit'] = df.apply(lambda x: revised_profit(x['revenue'], x['budget']),axis=1)
```

```
df.head()
```



imdb_id	belongs_to_collection	budget	genres	homepage	original_language	original_title	overview	popularity
tt2637294	[[{'id': 313576, 'name': 'Hot Tub Time Machine ...	14000000	[[{'id': 35, 'name': 'Comedy'}]]	NaN	en	Hot Tub Time Machine 2	When Lou, who has become the "father of the In...	6.57
tt0368933	[[{'id': 107674, 'name': 'The Princess Diaries ...	40000000	[[{'id': 35, 'name': 'Comedy'}, {'id': 18, 'name': 'Drama'}]]	NaN	en	The Princess Diaries 2: Royal Engagement	Mia Thermopolis is now a college graduate and ...	8.24
tt2582802	NaN	3300000	[[{'id': 18, 'name': 'Drama'}]]	http://sonyclassics.com/whiplash/	en	Whiplash	Under the direction of a ruthless instructor, ...	64.29
tt1821480	NaN	1200000	[[{'id': 53, 'name': 'Thriller'}, {'id': 18, 'name': 'Drama'}]]	http://kahaanithefilm.com/	hi	Kahaani	Vidya Bagchi (Vidya Balan) arrives in Kolkata ...	3.17
tt1380152	NaN	0	[[{'id': 28, 'name': 'Action'}, {'id': 53, 'name': 'Thriller'}]]	NaN	ko	마린보이	Marine Boy is the story of a former national s...	1.14

5 rows × 22 columns



✓ We can also sort our dataframe by some column values in either an ascending or descending order

```
# Sort values in descending order
df.sort_values('runtime', ascending=False)
```

	belongs_to_collection	budget	genres	homepage	original_language	original_title	overview	popularity
imdb_id								
tt1321865	NaN	18000000	[{'id': 80, 'name': 'Crime'}, {'id': 18, 'name': 'Drama'}]	NaN	en	Carlos	The story of Venezuelan revolutionary, Ilich R...	4.55807
tt0056937	NaN	31115000	[{'id': 18, 'name': 'Drama'}, {'id': 36, 'name': 'History'}]	NaN	en	Cleopatra	Historical epic. The triumphs and tragedy of t...	8.59123
tt0049833	NaN	13000000	[{'id': 18, 'name': 'Drama'}, {'id': 36, 'name': 'History'}]	NaN	en	The Ten Commandments	Escaping death, a Hebrew infant is raised in a...	11.14704
tt0080855	NaN	44000000	[{'id': 28, 'name': 'Action'}, {'id': 18, 'name': 'Drama'}]	NaN	en	Heaven's Gate	Harvard graduate James Averill (Kris Kristoffe...	4.39357
tt0279111	NaN	56000000	[{'id': 18, 'name': 'Drama'}, {'id': 36, 'name': 'History'}]	NaN	en	Gods and Generals	The film centers mostly around the personal an...	5.37456
...	...	...	...	...	...	...	...	...
tt1620549	[{'id': 106790, 'name': 'Hooked Collection', 'parent_id': 106790}]	3500000	[{'id': 28, 'name': 'Action'}, {'id': 878, 'name': 'Adventure'}]	NaN	ru	На Игре 2. Новьё Уровень	The continuation of the adventures of the hero...	0.88424
tt0768690	NaN	0	[{'id': 18, 'name': 'Drama'}, {'id': 35, 'name': 'History'}]	NaN	ru	А поутру они проснулись	NaN	0.40236
tt1133617	NaN	0	[{'id': 80, 'name': 'Crime'}, {'id': 18, 'name': 'Drama'}]	http://www.vt4.be/vermist	en	Vermist	Five members of the Missing Persons Task Force...	0.22923
tt1107828	NaN	6000000	[{'id': 18, 'name': 'Drama'}]	NaN	ru	Королёв	NaN	0.29229
tt0116485	NaN	0	[{'id': 35, 'name': 'Comedy'}]	NaN	de	Happy Weekend	NaN	0.00222

3000 rows × 22 columns

```
# We can also sort by the combination of two columns
df.sort_values(['status', 'new_profit'], ascending=[False, True]).head()
```

	belongs_to_collection		budget	genres	homepage	original_language	original_title	overview	popularity	productio
imdb_id										
tt0145394	NaN	891000	[[{'id': 99, 'name': 'Documentary'}]]	NaN	en	A Place Called Chiapas	On January 1, 1994, the Zapatista National Lib...	0.179246		
tt1860180	NaN	0	[[{'id': 35, 'name': 'Comedy'}]]	NaN	en	Billy Gardell: Halftime	Billy Gardell, the star of CBS' hit comedy "Mi...	0.007877		
tt0120254	[[{'id': 144200, 'name': 'Swan Princess Series'...]	0	[[{'id': 14, 'name': 'Fantasy'}, {'id': 16, 'na...	NaN	en	The Swan Princess: Escape from Castle Mountain	The first wedding anniversary of Princess Odet...	5.121309		
tt0091024	NaN	0	[[{'id': 18, 'name': 'Drama'}, {'id': 53, 'name...	NaN	en	Extremities	A woman escapes from the man who is about to r...	10.273979		[[{'n
tt2239822	NaN	197471676	[[{'id': 12, 'name': 'Adventure'}, {'id': 878, ...]	NaN	en	Valerian and the City of a Thousand Planets	In the 28th century, Valerian and Laureline ar...	15.262706		[[{'name':

5 rows × 22 columns

```
df.info()

<class 'pandas.core.frame.DataFrame'>
Index: 3000 entries, tt2637294 to tt1600195
Data columns (total 22 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   belongs_to_collection                 604 non-null    object
1   budget                               3000 non-null   int64
2   genres                               2993 non-null   object
3   homepage                             946 non-null    object
4   original_language                     3000 non-null   object
5   original_title                       3000 non-null   object
6   overview                             2992 non-null   object
7   popularity                           3000 non-null   float64
8   production_companies                 2844 non-null   object
9   production_countries                 2945 non-null   object
10  release_date                         3000 non-null   object
11  runtime                             2998 non-null   float64
12  spoken_languages                     2980 non-null   object
13  status                               3000 non-null   object
14  tagline                             2403 non-null   object
15  title                                3000 non-null   object
16  Keywords                             2724 non-null   object
17  cast                                 2987 non-null   object
18  revenue                              2998 non-null   float64
19  movie_profit                         3000 non-null   int64
20  Capitalized_title                    3000 non-null   object
21  new_profit                           2998 non-null   float64
dtypes: float64(4), int64(2), object(16)
memory usage: 603.6+ KB
```

```
df.info()

<class 'pandas.core.frame.DataFrame'>
Index: 3000 entries, tt2637294 to tt1600195
Data columns (total 24 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   id                                    3000 non-null   int64
1   belongs_to_collection                 604 non-null    object
```